TEXTBOOK EXERCISES

BPIQ97

I. Choose the correct answer.

1. Number of valence electrons in carbon is
   a) 2 b) 4 c) 3 d) 5

2. Sodium having atomic number 11, is ready to _____ electron/ electrons to attain the nearest noble gas electronic configuration.
   a) gain one b) gain two
   c) lose one d) lose two

3. The element that would form anion by gaining electrons in a chemical reaction is _____
   a) potassium b) calcium
   c) fluorine d) iron

4. Bond formed between a metal and non metal atom is usually _____
   a) ionic bond b) covalent bond
   c) coordinate bond

5. _____ compounds have high melting and boiling points.
   a) Covalent b) Coordinate c) Ionic

6. Covalent bond is formed by _____
   a) transfer of electrons
   b) sharing of electrons
   c) sharing a pair of electrons

7. Oxidising agents are also called as _____ because they remove electrons form other substances.
   a) electron donors b) electron acceptors

8. Elements with stable electronic configurations have eight electrons in their valence shell. They are ___
   a) halogens b) metals
   c) noble gases d) non metals

II. Answer briefly.

1. How do atoms attain Noble gas electronic configuration?

2. NaCl is insoluble in carbon tetrachloride but soluble in water. Give reason.

3. Explain Octet rule with an example.

4. Write a note on different types on bonds.

5. Correct the wrong statements.
   a. Ionic compounds dissolve in non polar solvents.
   b. Covalent compounds conduct electricity in molten or solution state.

6. Complete the table give below.

Element	Atomic number	Electron distribution	Valence electrons	Lewis dot structure
Lithium	3			
Boron	5			
Oxygen	8			

7. Draw the electron distribution diagram for the formation of Carbon di oxide (CO_2) molecule.

8. Fill in the following table according to the type of bonds formed in the given molecule. $CaCl_2$, H_2O, CaO, CO, KBr, HCl, CCl_4, HF, CO_2, Al_2Cl_6

Ionic bond	Covalent bond	Coordinate covalent bond

IX_SCI_EM_Unit-13_CHE.indd 159 12-02-2020 14:19:22